

The Hong Kong Polytechnic University
Department of Electronic and Information Engineering

EIE2111 Mini-Project

(Due date: Check the course information)

Objective

This mini-project is to design and implement a simple game, 2048.

Introduction

Please visit the following web site to understand the game and know the rules:

<https://play2048.co/>

Two students form a team and a team is required to create the game with GUI in this project.

Overall Design

Part I – Report and Presentation (10 marks)

Your report should discuss the following items in detail:

- Cover page, table of contents, page number, reference page (Show your full name, your student ID, your partner's name and your partner's student ID at the front page)
- Team work description (Write one paragraph or a table to describe the work allocation)
- Work schedule
- Class description
- Test cases for all basic and additional features with snapshots
- Conclusion

Your presentation should present and demonstrate all key and bonus features within 5 to 10 minutes in English. Moreover, you should play your games by yourself to demonstrate all features. Please register your group in Blackboard before the end of week 3. The schedule will be posted to Blackboard later.

Part II – Programming Work (90 marks)

By using the Microsoft Visual Studio, design and implement the simple game.

Basic requirement (60 marks)

- A user can create an account and log in his/her account (without passwords).
- The games can be played by your mouse in graphical mode. You should create four buttons (arrow keys) for the player to move the tiles. No animation is required. It means just show the result after the player presses a key.
- Show the best five records among all players at the end of a game.

Additional Features (30 marks, 10 marks for each feature, at most 3 additional features)

1. Game Mode
 - S1: The game is finished after the player scores more than a specified total score (e.g., 350).
 - S2: The game is finished after the playing time is over a specified period (e.g., 40 seconds).
 - S3: The game is finished after the player has taken more than a specified number of moves (e.g., 15 moves).
2. Effect Mode
 - E1: Sound effect (e.g., make a new high tile, miss)
 - E2: Animation effect (see the change of some tiles)
3. Special Characters
 - C1: A special character to remove a tile when it hits the tile.
 - C2: A special character to increase the value of a tile four times (e.g., tile 2 is changed to tile 8) when it hits the tile.
 - C3: A special character to extend the game by increasing its limits (i.e., the specified total score, the specified period or the specified number of moves).

We will assign one feature from each group (total three features) to each student team. The assignment of additional features will be posted to Blackboard after Week 4.

Instructions

1. You should zip all your programs and your report into one single file and then submit to Blackboard. Make sure you program code can be viewed and runnable again in our EIE laboratories.
2. The length of the written report should be not less than **SIX** pages excluding the cover, reference and table of content pages.
3. Every student must submit his/her report through Blackboard. It means a team of two students should submit two reports through their accounts in Blackboard respectively and it is expected that two reports are identical. You will score **NO MARKS** if you do not submit your report to Blackboard.
4. It is compulsory to use a word processing package to type your report. The programming work is the most important. However, you should also try your best to write clear, correct and good sentences in your report.

*Lawrence Cheung
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